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A Variant Fisher and Jaikuamr Algorithm to Solve Capacitated Vehicle Routing Problem

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***Abstract*— Vehicle Routing Problem (VRP) is a real life constraint satisfaction problem to find minimal travel distances of vehicles to serve customers. Capacitated CVRP (CVRP) is the simplest form of VRP considering vehicle capacity constraint. Constructive and clustering are the two popular approaches to solve CVRP. A constructive approach creates routes and attempts to minimize the cost at the same time. On the other hand, a clustering based method first assigns nodes into vehicle wise cluster and then generates route for each vehicle. Route generation is a traveling sales man problem (TSP) and any TSP optimization method is useful for this purpose. Fisher and Jaikumar algorithm is a well-known cluster based method which creates clusters with a geometric method partitioning the customer plane into equal angular cones where the total cones are equal to the number of vehicles. This study investigates a variant of Fisher and Jaikumar algorithm owning to solve CVRP efficiently. In the proposed method clusters form such a way that each cluster cone contains equal number of nodes. The initial clusters formation in a method are subjected to route optimization. Three prominent metaheuristic methods (i.e., genetic algorithm, ant colony optimization, particle swarm optimization) for TSP are considered for route optimization. The efficiency of the proposed method is identified while solving benchmark CVRPs. Experimental result reveals that variant Fisher and Jaikumar with particle swarm optimization is an effective method in solving CVRP.**

Keywords—Capacitated Vehicle Routing Problem; Fisher and Jaikumar algorithm, Genetic Algorithm, Ant Colony Optimization and Particle Swarm Optimization.

I. INTRODUCTION

The Vehicle Routing Problem (VRP) was first introduced by Dantzig and Ramser in [1] as "Track Dispatching Problem" can be described as the problem of designing optimal delivery routes from one or several depots to a number of geographically scattered cities or customers while maintaining some constraints. The Capacitated VRP (CVRP) is its most common form with an additional constraint of fixed vehicle carrying capacity [2]. The total demands of the customers assigned to a vehicle cannot exceed vehicle capacity. In general, CVRP considers equal capacities for all the vehicles; although vehicles with different capacities can be used. In CVRP, all customers have known demands and known locations for the delivery. The delivery for a customer cannot be split. In other words, the demand of a customer must be satisfied via only one visit. Goods are delivered from a single depot and the vehicles return to the depot after serving their assigned customers. The delivery or unloading time may or may not be considered. The objective is to minimize the total travelling distance for all vehicles.

Various ways of solving the CVRP have been investigated. Nearly all of them are heuristics because no exact algorithm can be guaranteed to find optimal tours within reasonable computing time when the number of nodes is large. A heuristic technique does not explore the entire search space; rather it tries to find an optimal solution based on the available information of the problem. Various constructive and clustering heuristic methods are investigated for CVRP. Constructive methods build a feasible solution gradually while keeping an eye on solution cost, but it may not contain an improvement or optimization phase. Some well-known algorithms of constructive methods are - Clarke and Wright's Savings Algorithm, Matching Based Algorithm and Multi-route Improvement Heuristics. On the other hand, in clustering method, the solution of the problem is divided into two phases; first grouping or clustering of customers maintaining given constraints and then generating optimal routes. Clustering based methods also categorized as 2-phase method: cluster first, route second method. In first phase clustering algorithm generates clusters of customers; second phase uses optimization technique to find optimum routes for each cluster. Sweep algorithm, Fisher and Jaikumar algorithm, Petal algorithm and Taillard's algorithm are some popular clustering algorithms.

Clarke and Wright's Savings algorithm [3] is the popular constructive algorithm and is based on savings concept. The savings distance for all pairs of customers are calculated and all pairs of customer points are sorted in descending order of the savings. Subsequently, from the top of the sorted list of point pairs one pair of points is considered at a time and they are inserted into routes. A number of studies based on Clarke and Wright savings algorithm are available for solving real life CVRPs such as solving A-VRP benchmark [4], in Dutch consultancy farm [5], optimization of transport planning of Tesco Prešov distribution center in Slovakia [6].

Sweep algorithm is the most studied cluster based 2-phase algorithm for CVRPs. It creates clusters of customers based on

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their angular position [7]. Initially clusters are formed by sweeping the nodes according to their polar angle (increasing or decreasing order). Sweeping halts when vehicle capacity constraint is violated, finishes clustering for a single vehicle and resumes for another vehicle. The capability of Sweep in solving the CVRP for public transport was investigated in [8]; Newspaper delivery in Bangkok in [9] and solving Augerat benchmark CVRPs in [10].

The Fisher and Jaikumar algorithm [11, 12] is also well known 2-phase algorithm for CVRPs. It creates clusters with a geometric method partitioning the customer plane into several cones where the cone number is equal to the vehicle number. Seed nodes are chosen from the cones and insertion cost is calculated. The customers are thus inserted into routes according to their increasing insertion cost. Once the clusters have been formed, then any Traveling Salesman (TSP) optimizing algorithm can be used to obtain the optimal travelled cost.

This study investigates a variant of Fisher and Jaikumar algorithm owning to solve CVRP efficiently. In the proposed method cluster form such a way that each cone contains equal number of nodes. The initial clusters formation in a method are subjected to route optimization. Three prominent metaheuristic methods (i.e., genetic algorithm, ant colony optimization, particle swarm optimization) for TSP are considered for route optimization. The efficiency of the proposed method is identified while solving benchmark CVRPs.

The outline of the paper is as follows; Section II explains standard Fisher and Jaikumar algorithm and proposed variant of it. Section III gives brief description of TSP optimization techniques considered in this study. Section IV is for experimental studies. At last, Section V gives a brief conclusion of the paper.

II. Variant Fisher and Jaikumar Algorithm to Solve CVRP

This section first describes standard Fisher and Jaikuamr (SFJ) Algorithm and explains its proposed variant.

A. Standard Fisher and Jaikuamr (SFJ) Algorithm

The Fisher and Jaikumar algorithm was introduced by Fisher and Jaikumar in 1981[11, 12]. It solves a Generalized Assignment Problem (GAP) to form the clusters. In GAP, there is a number of agents and a number of tasks. This problem is a generalization of the assignment problem in which both tasks and agents have a size. Moreover, the size of each task might vary from one agent to the other. Any agent can be assigned to perform any task, incurring some cost and profit that may vary depending on the agent-task assignment. Moreover, each agent has a budget and the sum of the costs of tasks assigned to it cannot exceed this budget. It is required to find an assignment in which all agents do not exceed their budget and total profit of the assignment is maximized. When GAP is applied for CVRP, the vehicles are considered agents, vehicle capacity is agent's budget, tasks are customers and profit is the optimal route cost. The assignment of customers to vehicles is done in such a way that vehicle capacity is not violated and the travelled distance is as less as possible.

SFJ proposed a geometric method to the partition the plane into total vehicle number of cones and seed vertices are chosen from the cones. Seeds are selected based on some criteria such as customers with maximum demand or most distant customer from origin. Insertion cost is calculated which is the cost of inserting customer within the route going back and forth between seed customer and depot. The customers are thus inserted into routes according to their increasing insertion cost. Once the clusters have been formed, then any TSP heuristics or optimizing algorithm can be used to obtain the optimal travelled cost.

Figure 1 demonstrates the route construction in SFJ. In the figure, 0 denotes the depot, S is seed customer and *i* is the customer being considered for insertion. Fig. 1(a) shows the back and forth route of seed customer and depot. Fig. 1(b) shows the route if customer *i* is visited in the way of returning from seed customer to depot. Now the travelled distance D_S of visiting the seed customer and returning to depot is,

$$D_S = D_{0S} + D_{S0} = 2D_S \tag{1}$$

and if customer *i* is visited on the way while visiting seed, then the distance D_i,

$$D_i = D_S + D_{Si} + D_{i0} \tag{2}$$

Substituting Eq. (2) from Eq. (1), insertion cost of customer *i,*

$$C_{ik} = D_i - D_S = D_{Si} + D_{i0} - D_S \tag{3}$$

The assignment of customers to vehicles is formulated as an integer problem and requires the following input:

C_{ik} = insertion cost of customer *i* and seed point *k*.

a_i = order size from customer *i*.

b_k = capacity of vehicle *k*.

Y_{ik} = 1 if customer *i* is assigned to vehicle *k*, 0 otherwise.

The integer program for assigning customers to vehicles is given by

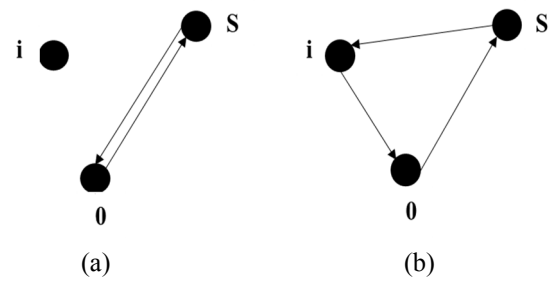


Fig. 1: demonstrates the route construction in SFJ. (a) Visiting seed customer and (b) Visiting both seed and customer *i*

$$Min \sum_{k=1}^{k} \sum_{i=1}^{n} Cik\ Yik \tag{4}$$

Subject to

$$\sum_{k=1}^{k} Yik = 1, \qquad i = 1,2, \dots . n$$

$$\sum_{i=1}^{n} ai\ Yik \leq bk, \qquad k = 1,2 \dots .. k$$

$$Yik = 1\ or\ 0\ for\ all\ i\ and\ k.$$

The first step of SFJ is to dividing the customers into several cones and then choosing a seed customer from each cone. Insertion cost is calculated for each customer with respect to each seed. At final step, the customers are assigned to vehicles according to the increasing order of insertion cost. In the SFJ algorithm, the customers are divided based on polar angle. The division starts from the node with the smallest angle, divides the angular plane in *K* equal cones (*K*= Number of vehicles) and ends at the largest angle. There may be no customer in some cones and for that case, the next cones are farther divided. Seed customer from each cone can be selected in one of the following two ways:

a. The most distant node from the origin and
b. The node with maximum demand.

Figure 2 demonstrates cluster creation and seed selection of a sample problem. In the figure the customers (N = 15) are divided into four cones (K = 4). If the smallest angle is 20^0 and the largest angle is 340^0 then each cone is of 80^0. From each cone, the farthest node from the origin is chosen as the seed

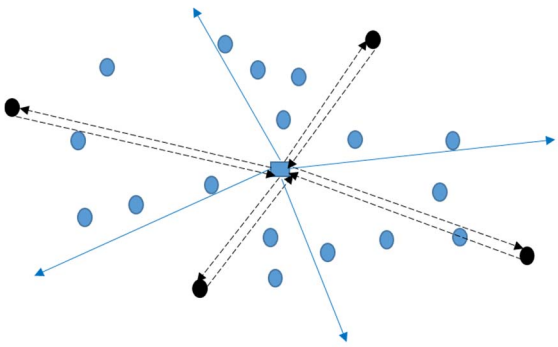


Fig. 2: Cone and seed customer selection in SFJ.

customer from that region.

B. Variant Fisher and Jaikuamr (VFJ) Algorithm

In the SFJ algorithm, the total nodes are divided into cones on angular basis. The problem with angular partition is that customer density may vary from cone to cone and there may be no customer in some particular cones for some instances. The original algorithm produces better solution for those instances where customers are evenly distributed around the depot. Hence, we are proposing a new way of forming the cones in which each cone may has average number of customer (total customer/total vehicle). It ensures equal customer distribution in all the partition and similar cluster size. The insertion cost calculation and assignment of customers are same as the original one.

Algorithm 1 shows the steps of the proposed VFJ algorithm. The plane dividing (Section 1.a) is the element of variant from SFJ. In VFJ cones holds equal number of nodes. Therefore, alteration of the portion with equal cone size will turn it SFJ.

***Algorithm 1:* Variant Fisher and Jaikumar (VFJ)**

1) ***Seed Selection***

a) **Divide the plane by the number of vehicle so that each cone holds equal number of nodes.**
b) Choose a customer from each cone with maximum demand or farthest from the origin and make that seed customer.
c) Assign a vehicle to each of the seed customer.

2) ***Insertion Cost Calculation***

Calculate insertion cost of each customer with respect to each seed using Eq. (3).

3) ***Assignment of Customers***

The customers are assigned to the vehicles according to their increasing order of insertion cost while maintaining vehicle capacity constraint.

III. ROUTE OPTIMIZATION

The route generation of individual vehicle is the final step to get CVRP solution with the clustered customer. The aim of it is to find the optimal route of a vehicle that represents the shortest path between all nodes in each cluster generated by the clustering algorithm. In this stage each cluster is an individual traveling salesman problem (TSP) and any TSP optimization method may useful. Three prominent TSP methods have been considered in this study which are genetic algorithm (GA), ant colony optimization (ACO), particle swarm optimization (PSO). GA is the prominent optimization method. ACO is very popular well-known Swarm Intelligence (SI) method for solving TSP. On the other hand, PSO is the most studied optimization in the recent time. Brief description of the methods are given to make the paper self-contained.

A. Genetic Algorithm (GA)

GA is one of the most popular search and optimization technique based on Darwin's Principle of *Natural Selection*. The common features of GA for TSP are: permutation encoding to present solution cluster, tournament selection, and Enhanced Edge recombination crossover and swap mutation [13].

B. Ant Colony Optimization (ACO)

ACO is inspired from ants' foraging behavior and is the prominent method for solving TSP. ACO is the first algorithm aiming to search for an optimal path in a graph, based on the behavior of ants seeking a path between their colony and a source of food. It considers population size as the number cities in a given problem and starts placing different ants in different cities. A particular ant consider next city to visit based on the visibility heuristic (i.e., inverse of distance) and intensity of the pheromone on the path. After the completion of a tour, each ant lays some pheromone on the path. Before pheromone deposit, pheromone evaporation of real ant is adopted by reducing pheromone of all the links by a fixed percentage. This behavior allows the artificial ants to forget bad choices made in the past. Finally, all the ants follow the same route after certain iteration. The detail description of ACO available in [14].

C. Particle Swarm Optimization (PSO)

PSO is a popular optimization method on metaphor of social behavior of flocks of birds or schools of fishes. In PSO, each particle represents a potential solution and moves to a new position (i.e., search a new point) at every iteration based on the calculated velocity [15]. The processes of iteration continue until the stopping criterion is reached. To solve TSP with PSO, each particle represents a complete tour as a feasible solution and velocity is a measure to update the tour for better solution. In this study most recent version of PSO for TSP, velocity tentative PSO (VTPSO) [16] is employed. VTPSO considers the calculated velocity as tentative velocity and conceives a measure called partial search to apply calculated velocity to update particle's position (i.e., TSP tour). VTPSO measures tours with portions of tentative velocity and conceive comparatively better new tour with a portion or full tentative velocity. The method is shown to perform well when tested on a suite of benchmark TSPs.

IV. EXPERIMENTAL STUDIES

This section describes the benchmark problems and experimental setup for conducting the experiment. Finally, the section represents experimental results and analysis.

A. Benchmark Data and Experimental Setup

In this study, a suite of benchmark problems named Augerat et al. (A-VRP) has been chosen [17]. Dataset includes the number of customer, number of vehicles available and each vehicle capacity. A customer is represented as a two dimensional coordinate, and demand of individual customer is also given. Number of customer varies from 32 to 80, total demand varies from 407 to 932, number of vehicle varies from 5 to 10 and capacity of individual is 100 for all the problems. The original data set is modified here to make the coordinate of the depot as (0, 0).

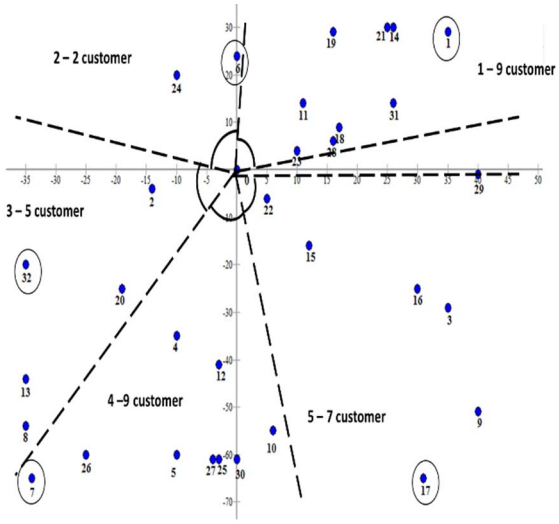
We strongly relied on an experimental methodology for configuring the TSP route optimization algorithms. For the fair comparison, the number of iteration was set at 100 for the algorithms. The number of ants in ACO was equal to the number of cities as it desired. The population size was 50 for GA, PSO and VTPSO. In ACO, *alpha* and *beta* were set to 1 and 3, respectively. The selected parameters are not optimal values, but considered for simplicity as well as for fairness in observation. The algorithms are implemented on Visual C++ of Visual Studio 2013. The experiments have been done on a PC (Intel Core i5-3470 CPU @ 3.20 GHz CPU, 4GB RAM) with Windows 7 OS.

B. Experimental Results and Analysis

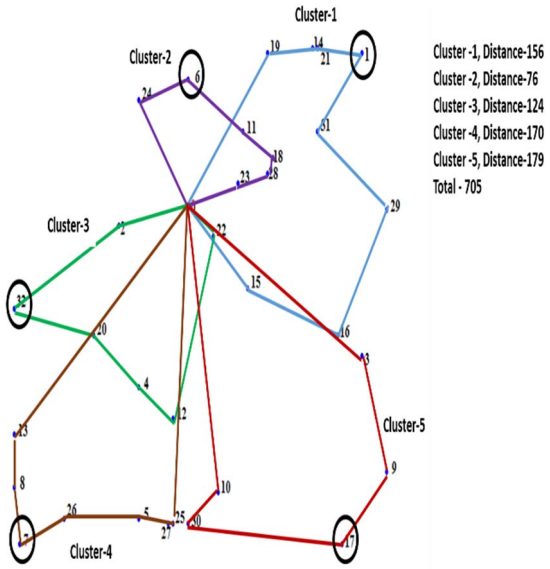
This section presents experimental results for the benchmark problems using SFJ and VFJ with route optimization with GA, ACO and PSO. First, a detailed observation for a problem with graphical representation is presented for better understanding and then experimental results are presented in tabular form.

Figure 3 shows graphical representation of the customers for standard SFJ using instance A-n33-k5 where the customers are divided into 5 polar cones with starting angle 22.56 and finishing angle 358.58. Seed customers from each cone are chosen as the most distant customer from depot and marked with circle. It is visible that the customers are not properly distributed among the cones; cone 1 and 4 contains 9 customers in each and cone 2 has only 2 customers. So during customer assignment, customers from cone 1 get inserted with seed 6 which belongs to cone 2. This results closer nearer customer insertion into different cluster and therefore outcome is larger cluster as show in Fig. 3(a). Thus partitioning the customers according to their polar angle affects the clustering and the final CVRP cost. Fig. 3(b) shows the routes generated by SFJ algorithm along with each route cost and total CVRP cost.

Figure 4 demonstrates the solving A-n33-k5 with proposed VFJ and PSO. VFJ partitions the customers based on their average number. In Fig.4 (a), each cone contains almost equal number of customers though they convers different angular space. Each of cone 1, 4 and 5 has 6 customers; and each of cone 2 and 3 has 7 customers. VFJ choses node 1, 14, 7, 30 and 17 as the seed customers whereas node 1, 6, 32, 7 and 17 was chosen in SFJ. This nodal division of the customers instead of angular division ensures that the clusters have almost equal number of customers and customers from the same cone are inserted into the same cluster. Therefore, better clusters are produced and CVRP cost is more optimized. Fig. 4(b) shows the routes generated using VFJ along with each route cost and total CVRP cost. It is also observed that the clusters are not intersecting for VFJ while SFJ clusters interests. Also the cluster size is smaller for VFJ in comparison with SFJ.

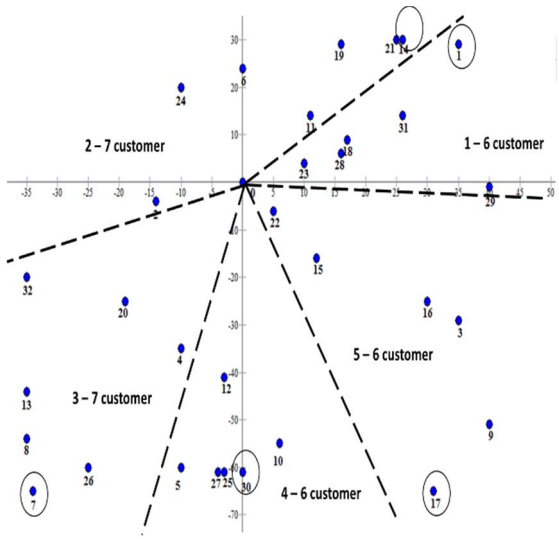


(a) Customer cones are created with equal angles in SFJ.

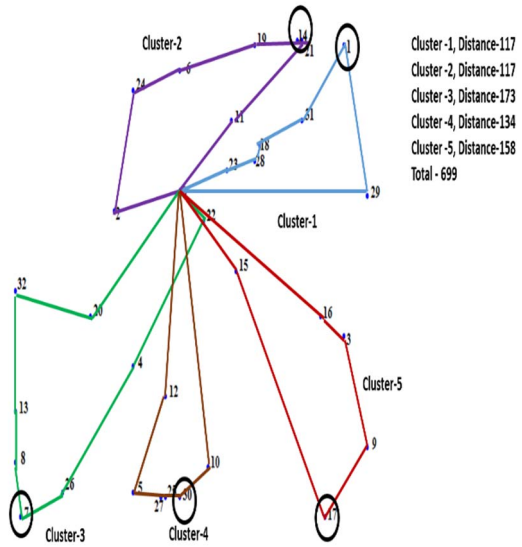


(b) CVRP solution with PSO.

Fig. 3: Solving A-n33-k5 using SFJ + PSO.



(a) Customer cones are created with equal nodes in VFJ



(b) CVRP solution with PSO.

Fig. 4: Solving A-n33-k5 using VFJ + PSO.

Tale 1 shows the CVRP result for SFJ and VFJ for GA, ACO and PSO. From the table, it can be observed that for the same instances and using same optimization technique, the route costs are different due to different approach of cluster/route generation. The best route found for each TSP algorithm is shown in underlined and best for each CVRP instance is also shown in bold.

From the table it is observed that proposed VFJ performed better than SFJ for 19 cases out of 27 while SFJ performed better for 8 cases. From the table it is also observed that for many instances same result is obtained for different optimization techniques. For A-n34-k5 and A-n39-k6, VFJ produced same route cost for all the three optimization techniques GA, ACO and PSO. This is due to the smaller size of the benchmark instances. It is also observed that GA is better than ACO; and PSO is the best for most of the instances**.** In case of optimization with PSO, VFJ outperformed SFJ for 17 cases out of 27. Moreover, VFJ with PSO achieved minimum average CVRP costs of 1150.78. From the table, it is clear that the proposed VFJ outperformed SFJ to get better CVRP solution. And VFJ with PSO is an effective CVRP solving method.

IV. CONCLUSIONS

TABLE I. SOLUTIONS OF A-VRP CVRPS THROUGH CLUSTERING WITH STANDARD FISHER AND JAIKUMAR (SFJ) AND VARIANT FISHER AND JAIKUMAR (VFJ) AND OPTIMIZING WITH GA , ACO AND PSO.

SL.	Problem	SFJ			VFJ		
		GA	ACO	PSO	GA	ACO	PSO
1	A-n32-k5	886	894	878	860	870	**844**
2	A-n33-k5	705	705	705	**699**	731	**699**
3	A-n33-k6	856	876	854	**804**	812	**804**
4	A-n34-k5	870	871	870	**841**	**841**	**841**
5	A-n36-k5	844	851	**828**	859	861	855
6	A-n37-k5	742	753	**726**	755	800	755
7	A-n37-k6	1016	1045	**1014**	1073	1076	1072
8	A-n38-k5	798	781	778	783	781	**776**
9	A-n39-k5	900	923	**894**	942	953	936
10	A-n39-k6	964	974	949	**865**	**865**	**865**
11	A-n44-k6	1162	1184	1159	**1018**	1026	**1018**
12	A-n45-k6	**1029**	1045	**1029**	1030	1070	1030
13	A-n45-k7	1216	1202	**1193**	1270	1326	1270
14	A-n46-k7	1040	1035	1019	**1008**	1049	**1008**
15	A-n48-k7	1186	1183	**1160**	1162	1190	**1160**
16	A-n53-k7	1251	1270	1242	1206	1236	1193
17	A-n54-k7	1194	1203	1194	1249	1277	**1243**
18	A-n55-k9	1283	1282	1274	**1189**	1203	**1189**
19	A-n60-k9	1756	1756	1756	1476	1484	**1462**
20	A-n61-k9	1141	1154	**1137**	1234	1228	1207
21	A-n62-k8	1664	1674	1655	1464	1493	**1460**
22	A-n63-k9	**1789**	1806	**1789**	1860	1904	1857
23	A-n63-k10	1499	1499	1499	1496	1511	**1474**
24	A-n64-k9	1613	1603	1572	1513	1560	**1474**
25	A-n65-k9	1376	1351	1346	1338	1386	**1335**
26	A-n69-k9	1359	1367	1340	1312	1302	**1301**
27	A-n80-k10	2056	2099	2019	1966	1984	**1943**
Average		1192.41	1199.48	1180.70	1158.22	1178.48	**1150.78**
Outperformance of VFJ over corresponding SFJ					18	12	17

Capacitated vehicle routing problem (CVRP) is a real life constraint satisfaction problem in which customers are optimally assigned to individual vehicles (considering their capacity) to keep total travel distance of the vehicles as minimum as possible while serving customers. This study investigated a clustering based CVRP solving method in which a variant of Fisher and Jaikuamr algorithm is considered for clustering the nodes and three prominent TSP methods (i.e., GA, ACO and PSO) are employed for optimizing routes of individual vehicles. The experiment results reveal that variant Fisher and Jaikuamr with PSO is an efficient method for solving CVRPs.

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